

1. Administrative & Study Identification

Full Study Title A Randomized, Double-Blind, Phase 3 Study of Pembrolizumab/Vibostolimab Coformulation (MK-7684A) in Combination with Chemotherapy Versus Pembrolizumab Plus Chemotherapy as First Line Treatment for Participants with Metastatic Non-Small Cell Lung Cancer (MK-7684A-007/KEYVIBE-007)
Short Title/Acronym KEYVIBE-007 (MK-7684A-007) **Protocol Number** MK-7684A-007 **NCT Number** NCT05226598 **Trial Status** ACTIVE_NOT_RECRUITING (Active but not currently recruiting)
Sponsor Name Merck Sharp & Dohme LLC (MSD) / Merck
Investigational Product MK-7684A (pembrolizumab/vibostolimab coformulation) **Phase of Development** Phase III **Medical Monitor** Clinical Director, Merck Sharp & Dohme LLC

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine if pembrolizumab/vibostolimab (MK-7684A) in combination with chemotherapy is superior to pembrolizumab in combination with chemotherapy with respect to overall survival (OS) in participants with programmed cell death-ligand 1 (PD-L1) tumor proportion score (TPS) $\geq 1\%$. **Secondary Objectives:** To evaluate OS in all participants; Progression-Free Survival (PFS); Objective Response Rate (ORR); Duration of Response (DOR) in TPS $\geq 1\%$ and all participants; safety and tolerability; and change in patient-reported outcomes (PROs).

2.2 Background & Rationale Lung cancer is frequently diagnosed at an advanced stage, making systemic treatment challenging. TIGIT (T-cell immunoglobulin and ITIM domain) is an inhibitory receptor expressed on immune cells that blunts anti-tumor responses. Vibostolimab is an investigational humanized anti-TIGIT antibody. Preclinical and early-phase data suggested that combined blockade of the PD-1 (with pembrolizumab) and TIGIT (with vibostolimab) pathways could enhance T-cell activation and tumor cell clearance. The KEYVIBE-007 study aims to assess this synergistic approach as a first-line treatment for metastatic NSCLC.

(Note: Effective as of Amendment 5, participants receiving coformulation of pembrolizumab/vibostolimab plus chemotherapy will be transitioned to standard of care [SOC, pembrolizumab plus chemotherapy]. Participants with access to approved SOC should be considered for discontinuation from the study. Those benefiting from pembrolizumab plus chemotherapy, but unable to access it as SOC outside the study, may continue on study and receive

treatment with pembrolizumab plus chemotherapy until discontinuation criteria are met.)

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Pembrolizumab + Chemotherapy) **Blinding:** Double-blind **Randomization:** 1:1, Stratified by ECOG Performance Status (0 vs 1), tumor histology (squamous vs nonsquamous), PD-L1 expression (TPS $\leq 50\%$ vs $\geq 50\%$), and geographic region (East Asia vs North America/Western Europe vs rest of world).

3.2 Planned Enrollment & Duration Target N: Approximately 739 participants **Number of Sites:** Global (Five EU countries, UK, US, and others) **Estimated Study Start:** March 2022 **Estimated Primary Completion:** November 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically or cytologically confirmed diagnosis of Stage IV squamous or non-squamous Non-Small Cell Lung Cancer (NSCLC). 3. Has not received prior systemic treatment for metastatic NSCLC. 4. Has measurable disease based on RECIST 1.1, as determined by the local site assessment. 5. Has a life expectancy of at least 3 months. 6. Males: Use contraception unless confirmed to be azoospermic; Females: Women of childbearing potential use highly effective contraceptive method.

4.2 Exclusion Criteria 1. Prior systemic anticancer therapy for metastatic disease. 2. Known additional malignancy that is progressing or has required active treatment within the past 3 years. 3. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 4. Severe hypersensitivity to MK-7684, MK-7684A, pembrolizumab, chemotherapy components, and/or any of its excipients. 5. Active autoimmune disease that has required systemic treatment in past 2 years (except replacement therapy), or a condition requiring systemic immunosuppressive therapy. 6. Received a live or live attenuated vaccine within 30 days before the first dose of study intervention.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MK-7684A (pembrolizumab 200 mg / vibostolimab 200 mg coformulation) administered on Day 1 of each 3-week cycle, plus standard-of-care chemotherapy (e.g., cisplatin [Platinol]). **Route of Administration:** Intravenous (IV) infusion **Duration of Treatment:** Cycles are repeated every 3 weeks until disease progression, unacceptable toxicity, or completion of the maximum allowed

treatment duration (e.g., up to 35 cycles for the immunotherapy component).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care treatments (e.g., antiemetics for chemotherapy, pain management). **Prohibited:** Concurrent investigational agents, other anti-cancer immunotherapies, and live/live-attenuated vaccines.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Tumor Imaging, PD-L1 testing | | **Baseline** | Day 0 | Randomization, First Dose of MK-7684A or Pembrolizumab + Chemo | | **Interim** | Weeks 6, 12, 18 | Safety Labs, Tumor Imaging, PRO Collection. Followed by Q9W through Week 63, then Q12W. | | **End of Study** | At progression | Final Safety Assessments, Transition to Follow-up, Survival Tracking |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Overall Survival (OS), defined as the time from the date of randomization to death due to any cause, in participants with PD-L1 TPS $\geq 1\%$. **Measurement Method:** Clinical status tracking and survival follow-up. Calculated using the product-limit (Kaplan-Meier) method for censored data.

7.2 Secondary & Exploratory Endpoints - Overall Survival (OS) in all participants. - Progression-Free Survival (PFS) per RECIST v1.1 assessed by Blinded Independent Central Review (BICR). - Objective Response Rate (ORR) and Duration of Response (DOR) per RECIST v1.1 by BICR. - Change from baseline and time to true deterioration in patient-reported outcomes (PROs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection and grading according to National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) v5.0. **Serious Adverse Events (SAEs):** Routine expedited reporting to sponsor within 24 hours. Immune-related AEs receive special monitoring. **Data Monitoring Committee (DMC):** Independent, external Data Monitoring Committee oversees unblinded safety and efficacy data throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Population for efficacy outcomes; As-Treated (Safety) Set for adverse events.

Sample Size Justification: Approximately 739 targeted participants randomized 1:1, powered ($\alpha = 0.05$, expected power) to detect a statistically significant difference in OS between the investigational and active comparator arms for the TPS $\geq 1\%$ population. **Handling of Missing Data:** Standard survival analysis algorithms (e.g., Kaplan-Meier survival estimates, with censoring for patients lost to follow-up or alive at the data cutoff).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Periodic on-site and remote monitoring visits to ensure subject safety, protocol adherence, and integrity of the clinical data. **Audit Trail:** Robust electronic Case Report Form (eCRF) documentation, database locking procedures, and query management systems.

11. Study Identifiers & Governance

Primary ID: MK-7684A-007 **Posting ID:** 1096279879358278590 **Registry Number:** NCT05226598 **Sponsor/Lead Organization:** Merck / Merck Sharp & Dohme LLC **Study Title:** Study of Pembrolizumab/Vibostolimab Coformulation (MK-7684A) in Combination With Chemotherapy Versus Pembrolizumab Plus Chemotherapy in Participants With Metastatic Non-Small Cell Lung Cancer (MK-7684A-007/KEYVIBE-007)

12. Clinical Framework (Oncology Specific)

Primary Condition: Metastatic Non-Small Cell Lung Cancer (NSCLC)

Diagnosis Threshold: Histologically or cytologically confirmed Stage IV disease **Medical Methodology:** Combined Immuno-Oncology and Cytotoxic Chemotherapy (including cisplatin/Platinol)

Optimization Target: Blockade of the PD-1/PD-L1 and TIGIT signaling pathways to overcome tumor immune evasion

13. Study Procedures & Methodology

Management Pathway: Systemic first-line intervention for advanced NSCLC. **Education Protocol (HCP):** Site initiation and continuous training regarding protocol requirements and management of immune-related adverse events. **Education Protocol (Patient):**

Comprehensive Informed Consent process and patient diaries/questionnaires. **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for all tumor imaging and routine sponsor auditing.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up continues until disease progression, death, or study withdrawal. **Onsite Visit Cadence:** Every 3 weeks (Q3W) for dosing; tumor imaging at Weeks 6, 12, 18, then every 9 weeks (Q9W) through Week 63, and every 12 weeks (Q12W) thereafter. **Remote Monitoring Frequency:** Regular outreach during the survival follow-up phase post-progression. **Checklist Review Interval:** Validated PRO instruments (EORTC QLQ-LC13, NSCLC-SAQ) assessed at predetermined clinical visits.

15. Sign-off & Approval

Role	Name	Signature	Date	:	---	:	---	:	---	:	---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]	:	---	:	---	:	---	:	---
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]	:	---	:	---	:	---	:	---
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]	:	---	:	---	:	---	:	---